Name: ____________________

WORKSHEET

Multiplying matrices

Nothing can stop the man with the right mental attitude from achieving his goal; . . .

14 10 15 12 2 14 9   10 14   8 11 4 15 12

3 11 14   12 8 5 7   15 12 8   16 11 14

13 2 15 12   15 12 8   13 4 10 14 9

16 8 14 15 11 5   11 15 15 2 15 1 6 8

Thomas Jefferson

Questions

Is it possible to multiply matrices of the following orders:

1 **AB**, where **A** is a 3 × 3 matrix and **B** is a 3 × 3 matrix.

2 **CD**, where **C** is a 5 × 2 matrix and **D** is a 2 × 5 matrix.

3 **DC**, where **D** and **C** are as above.

4 **EF**, where **E** is a 3 × 4 matrix and **F** is a 5 × 4 matrix.

What is the order of **MN** if:

5 **M** is a 5 × 3 matrix and **N** is a 3 × 2 matrix?

6 **M** is a 4 × 4 matrix and **N** is a 4 × 1 matrix?

7 **M** is a 7 × 2 matrix and **N** is a 2 × 5 matrix?

Solutions

A $\begin{bmatrix} -23 & 13 \\ 29 & 33 \end{bmatrix}$

D 4 × 1

E $\begin{bmatrix} 1 & 6 \\ 29 & 12 \end{bmatrix}$

G $\begin{bmatrix} 17 \\ 17 \end{bmatrix}$

H $\begin{bmatrix} 2 \\ 18 \\ -34 \\ -9 \end{bmatrix}$

Calculate the following:

8 $\begin{bmatrix} 5 & 2 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} -3 & 0 \\ 8 & 3 \end{bmatrix}$

9 $\begin{bmatrix} 4 & -9 & -2 \\ 7 & 0 & 6 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix}$

10 $\begin{bmatrix} 9 & -2 \\ 4 & 7 \end{bmatrix}\begin{bmatrix} -3 & 1 \\ 1 & 5 \end{bmatrix}$

11 $\begin{bmatrix} -3 & 1 \\ 1 & 5 \end{bmatrix}\begin{bmatrix} 9 & -2 \\ 4 & 7 \end{bmatrix}$

12 $\begin{bmatrix} 2 & 6 \\ 0 & -9 \\ -4 & 3 \\ 1 & 8 \end{bmatrix}\begin{bmatrix} 7 \\ -2 \end{bmatrix}$

13 $\begin{bmatrix} 5 & -2 & 3 \\ 0 & 9 & -4 \end{bmatrix}\begin{bmatrix} 6 & -1 \\ 7 & 8 \\ -3 & 5 \end{bmatrix}$

14 $\begin{bmatrix} 3 & 5 & -9 \\ 7 & -4 & 2 \end{bmatrix}\begin{bmatrix} 0 & 6 & -1 \\ 4 & -8 & -7 \end{bmatrix}$

15 $\begin{bmatrix} 9 & -2 \\ -3 & 5 \end{bmatrix}\begin{bmatrix} 0 & 7 \\ 4 & 1 \end{bmatrix}$

16 $\begin{bmatrix} 2 & -2 & 5 \\ 1 & 0 & 7 \\ 8 & 4 & -3 \end{bmatrix}\begin{bmatrix} -6 & 1 & 3 \\ 4 & 7 & 0 \\ -2 & 8 & -1 \end{bmatrix}$

Solutions

L 5 × 2

M $\begin{bmatrix} -30 & 28 & 1 \\ -20 & 57 & -4 \\ -26 & 12 & 27 \end{bmatrix}$

N Not possible.

O $\begin{bmatrix} -29 & -1 \\ -5 & 39 \end{bmatrix}$

P 7 × 5

T $\begin{bmatrix} -8 & 61 \\ 20 & -16 \end{bmatrix}$

W $\begin{bmatrix} 7 & -6 \\ 75 & 52 \end{bmatrix}$

Solutions	
YES	NO
U	F
I	K
C	S
B	R